Forces: Vectors

Questions 1–23

Question 1. Two forces $\mathbf{P}$ and $\mathbf{Q}$ act on a particle. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively. Find the magnitude and direction (anticlockwise from the positive x-axis) of the resultant force in each case, giving your answers to 3 significant figures:

a) $\mathbf{P} = (2\mathbf{i} - \mathbf{j})$ N and $\mathbf{Q} = (-\mathbf{i} + 4\mathbf{j})$ N
b) $\mathbf{P} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}$ N
c) $\mathbf{P} = (3\mathbf{i} + \mathbf{j})$ N and $\mathbf{Q} = (-5\mathbf{i} + 8\mathbf{j})$ N
d) $\mathbf{P} = \begin{pmatrix} -5 \\ -9 \end{pmatrix}$ N and $\mathbf{Q} = 4\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ N
e) $\mathbf{P} = (-0.5\mathbf{i} + 2\mathbf{j})$ N and $\mathbf{Q} = 3(\mathbf{i} - 6\mathbf{j})$ N

Question 2. Two forces $\mathbf{P}$ and $\mathbf{Q}$ act on a particle. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively. Find the magnitude and the bearing of the resultant force in each case:

a) $\mathbf{P} = (5\mathbf{i} + 3\mathbf{j})$ N and $\mathbf{Q} = (-\mathbf{j})$ N
b) $\mathbf{P} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ N
c) $\mathbf{P} = (2\mathbf{i} - \mathbf{j})$ N and $\mathbf{Q} = (3\mathbf{i} - 11\mathbf{j})$ N
d) $\mathbf{P} = \begin{pmatrix} 5 \\ -1.2 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} -6 \\ 1 \end{pmatrix}$ N
e) $\mathbf{P} = (1.25\mathbf{i} - 0.75\mathbf{j})$ N and $\mathbf{Q} = (-2.25\mathbf{i} - 1.75\mathbf{j})$ N

Question 3. The force $\binom{k}{5}$ N where k is a positive constant, has a magnitude of $\sqrt{29}$ N. Find the value of k

Question 4. The force $(2k\mathbf{i} - 2k\mathbf{j})$ N where k is a constant, has a magnitude of 12 N and a bearing of 315°. Find the value of k

Question 5. Two forces **P** and **Q** act on a particle. The vectors **i** and **j** are unit vectors due east and north respectively. Find a third force **R** such that the three forces are in equilibrium:

a) $\mathbf{P} = (-2\mathbf{i} + \mathbf{j})$ N and $\mathbf{Q} = (-3\mathbf{i} + 11\mathbf{j})$ N
b) $\mathbf{P} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} -4 \\ \sqrt{3} \end{pmatrix}$ N
c) $\mathbf{P} = (\sqrt{2}\mathbf{i} - \mathbf{j})$ N and $\mathbf{Q} = (-3\mathbf{j})$ N
d) $\mathbf{P} = \begin{pmatrix} k \\ 1 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} -k \\ 1 \end{pmatrix}$ N where k is a constant.
e) $\mathbf{P} = (a\mathbf{i} + b\mathbf{j})$ N and $\mathbf{Q} = (-a\mathbf{i} + 2b\mathbf{j})$ N where a and b are constants.
f) $\mathbf{P} = \begin{pmatrix} 2a \\ 6a \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} b \\ 6b \end{pmatrix}$ N where a and b are constants.

Question 6. The forces $\begin{pmatrix} a \\ 3 \end{pmatrix}$ N, $\begin{pmatrix} 2b \\ -a \end{pmatrix}$ N and $\begin{pmatrix} -6b \\ b \end{pmatrix}$ N act on an object which is in equilibrium. Find the values of a and b

Question 7. Three forces, $(6\mathbf{i} + 10\mathbf{j})$ N, $(-3.5\mathbf{i} - 12\mathbf{j})$ N and $(-2.5\mathbf{i})$ N act on a particle. It begins at rest at the origin. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively.

a) Find the resultant force in the form $(a\mathbf{i} + b\mathbf{j})$ N where a and b are constants to be found.
b) Find the magnitude and the bearing of the resultant force.
c) Describe the motion of the particle.
d) State the additional force that would be needed for the particle to instead be at rest.

Question 8. Three forces, $(12\mathbf{i} + 11\mathbf{j})$ N, $(3\mathbf{i} - 15\mathbf{j})$ N and $(a\mathbf{i} + b\mathbf{j})$ N act on a particle. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively. State the values of a and b and b such that the particle is moving due East.

Question 9. Two forces $\mathbf{P}$ and $\mathbf{Q}$ act on a particle. Find the angle between the two forces:

a) $\mathbf{P} = (3\mathbf{i} + 4\mathbf{j})$ N and $\mathbf{Q} = (5\mathbf{i} + 12\mathbf{j})$ N

b) $\mathbf{P} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ N and $\mathbf{Q} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$ N

Question 10. Two forces are given by $\begin{pmatrix} 1 \\ -7 \end{pmatrix}$ N and $\begin{pmatrix} k \\ 3k \end{pmatrix}$ N, where k is a positive constant. Given that the resultant of these two forces is parallel to $\begin{pmatrix} -4 \\ 7 \end{pmatrix}$, find the value of k

Question 11. Three forces, $\begin{pmatrix}1\\2\end{pmatrix}$ N, $\begin{pmatrix}-3\\8\end{pmatrix}$ N and $\begin{pmatrix}a\\b\end{pmatrix}$ N act on a particle. The vectors **i** and **j** are unit vectors due east and north respectively.

a) State the values of a and b such that the particle is moving due North.

It is given that $a = 8$

b) Find the value of b such that the particle is moving due North-East.

Question 12. Find the unit vector of each of the following vectors:

a) $\begin{pmatrix}3\\4\end{pmatrix}$

b) $3\mathbf{i} - \mathbf{j}$

c) $\begin{pmatrix}-1\\4\end{pmatrix}$

d) $-8\mathbf{i} - 6\mathbf{j}$

Question 13. The force $\mathbf{F}$ is parallel to $\overrightarrow{AB}$. Given the coordinates of A and B and the magnitude of $\mathbf{F}$, find $\mathbf{F}$ in each case:

a) $A(0,0),\ B(-3,-3),\ |\mathbf{F}| = 5$ N
b) $A(0,-2),\ B(3,0),\ |\mathbf{F}| = 4$ N
c) $A(1,2),\ B(5,6),\ |\mathbf{F}| = 8$ N

Question 14. Prerequisite knowledge for Q14-Q23: Constant Acceleration A resultant force of $(3\mathbf{i} + 4\mathbf{j})$ N is pushing against a particle of mass 0.5 kg. The particle is initially at rest.

a) Find the magnitude of the acceleration of the mass
b) Hence find the time it takes for the particle to move 80 metres.

Question 15. A resultant force of $\binom{1}{2}$ N is pushing against a particle of mass 2 kg. The particle is initially at rest.

a) Find the magnitude of the acceleration of the mass.

b) Hence find its speed of motion after 4 seconds.

Question 16. Two forces given by $(5\mathbf{i} + \mathbf{j})$ N and $(-2\mathbf{i} + 3\mathbf{j})$ N are pushing against a particle of mass 4 kg. The particle is initially at rest.

a) Find the magnitude of the acceleration of the mass.

b) Hence find the time it takes for the particle to move 10 metres.

Question 17. A force of $\begin{pmatrix} 5 \\ 11.6 \end{pmatrix}$ N is pulling on an object of mass 2kg which is falling under the influence of gravity, where $g = 9.8$ ms^{-2}. Find the magnitude of the acceleration of the object.

Question 18. An object of mass 2 kg is dropped. A force of $(4.5\mathbf{i} + 13.6\mathbf{j})$ N is pulling on the object as it falls under the influence of gravity, where $g = 9.8$ ms^{-2}.

a) Find the magnitude of the object's acceleration.

b) Another force of $(-4.5\mathbf{i} + 5.5\mathbf{j})$ N is applied to the object. Find the magnitude of the object's acceleration now.

Question 19. A resultant force **F** N parallel to $4\mathbf{i} - 3\mathbf{j}$ is pulling a box of mass 4 kg. The particle is initially at rest. It takes 2 seconds to travel 5 metres. Find the vector form of **F**

Question 20. A resultant force **F** N parallel to $8\mathbf{i} + 4\mathbf{j}$ is pulling a box of mass 3 kg. Initially it is moving at 1 ms^{-1} but reaches 4 ms^{-1} in 5 seconds. Find the vector form of **F**

Question 21. Two vectors $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ N and $\begin{pmatrix} p \\ q \end{pmatrix}$ N are acting on a box of mass 5 kg which is initially at rest. The box moves in the direction $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

a) Show that $p - 2q + 7 = 0$

Given that $p = 5$

b) Find the distance the box has travelled in 5 seconds.

Question 22. Two vectors $3\mathbf{i} - \mathbf{j}$ N and $p\mathbf{i} + q\mathbf{j}$ N are acting on a box of mass 3 kg which is initially at rest. The box moves in the direction $\mathbf{i} + 5\mathbf{j}$

a) Show that $5p - q + 16 = 0$

Given that $q = 6$

b) Find the speed of the box after 3 seconds.

Question 23. Two vectors $\begin{pmatrix}-1\\4\end{pmatrix}$ N and $\begin{pmatrix}p\\q\end{pmatrix}$ N are acting on a box of mass 2 kg which is initially at rest. The box moves in the direction $\begin{pmatrix}3\\4\end{pmatrix}$

a) Show that $4p - 3q - 16 = 0$

It begins at position A with a speed of 2 ms^{-1}. Given that $p = -2$

b) Find the speed of the box after 3 seconds.